



# FedBook-Sem

A Federated Social Layer for an Intelligent Online Bookstore

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# Motivation & Research Gap

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## The Problem

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- Book platforms are silos — Goodreads, Amazon, StoryGraph each keep their own social graph.
- Reader-generated annotations are locked to one product and lost on migration.
- Recommenders ignore the wider social conversation happening on Reddit, YouTube, Bluesky and Mastodon.
- There is no open specification for interoperable, scholarly-grade book annotations.

## Research Gap

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- Federated social protocols (ActivityPub) are proven for microblogging (Mastodon), but under-used for books.
- W3C Web Annotation is standardised for scholarly texts, rarely applied to trade-fiction reading UIs.
- Cross-platform reception is rarely folded into a single recommender score — most systems use only intrinsic ratings.
- No existing system combines all three in one interoperable stack.

## Research Questions

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- RQ1 — Can ActivityPub sustain a book-focused social network with W3C annotations as first-class activities?
- RQ2 — Does adding privacy-preserving cross-platform reception measurably improve recommendation quality (NDCG@10, MRR) over semantic or CF alone?
- RQ3 — Can learned rank fusion (LightGBM LambdaRank) combine heterogeneous signals better than a fixed-weight linear ensemble?

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# Contribution

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## Contributions

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- A federated book-social protocol layered on ActivityPub with W3C annotations as CREATE activities.
- A privacy-preserving ingestion pipeline that scores cross-platform sentiment without persisting raw mention text.
- A hybrid recommender combining semantic + CF + reception + graph signals, with an LTR fusion head.
- An open, reproducible dev environment (Docker Compose, seedable Neo4j, offline-cached models).
- A Streamlit evaluation dashboard exposing every recommender strategy side-by-side.

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# Methodology

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## Methodology — Design Science

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- Design Science Research Methodology (Peppers et al.) — iterative build-and-evaluate cycles.
- Six artefacts: federation protocol, annotation model, ingestion pipeline, recommender ensemble, dashboard, evaluation harness.
- Each artefact justified against literature, then evaluated with a mix of technical and human metrics.
- Reproducibility: single command (docker compose up -d) reconstructs the entire stack from source.

# Evaluation Plan

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## Recommender quality (offline)

- NDCG@5, NDCG@10, MRR, Recall@20
- Held-out split on goodbooks-10k
- Ablation: semantic-only vs. +reception vs. +CF vs. full LTR
- Coverage & catalogue diversity (Shannon entropy)

## System & user (mixed)

- Federation conformance vs. Mastodon reference server
- Annotation round-trip (W3C validator)
- 10-user think-aloud study, task-completion time
- Latency: P50 / P95 on /recommend/similar

## Ethical & Legal Considerations

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- Raw mention text from Reddit / YouTube / Bluesky / Mastodon is never persisted — only aggregated sentiment counts and external IDs for the deletion sweep.
- YouTube data honours the 30-day retention window (`purge_expired_receptions.py`).
- Reddit deletion sweep (`reddit_deletion_sweep.py`) removes counts for deleted posts.
- All external APIs are called with a project user-agent identifying the researcher.
- User study cleared by SLIIT research ethics; consent form + PIS on file.

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# System

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# Technology Stack

|                       |                                                                    |
|-----------------------|--------------------------------------------------------------------|
| <b>Federation</b>     | ActivityPub + WebFinger + HTTP Signatures (RSA-2048)               |
| <b>Annotations</b>    | W3C Web Annotation Data Model — TextQuote + TextPosition selectors |
| <b>Backend</b>        | Node.js 20 · Express · JWT · bcrypt · node-forge · multer · Bull   |
| <b>Frontend</b>       | React 18 · React Router v6 · Axios · custom dark design system     |
| <b>Graph DB</b>       | Neo4j 5.15 + APOC + Graph Data Science + native vector index       |
| <b>ML service</b>     | Python 3.11 · FastAPI · sentence-transformers · LightFM · LightGBM |
| <b>Sentiment</b>      | CardiffNLP twitter-roberta-base-sentiment-latest (RoBERTa)         |
| <b>Cross-encoder</b>  | BAAI/bge-reranker-base (~500 MB)                                   |
| <b>Dashboard</b>      | Streamlit — Recommender demo + Model registry                      |
| <b>Infrastructure</b> | Docker Compose · Redis 7 · PostgreSQL 16                           |

## Architecture at a Glance

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React FE → Express API → Neo4j graph, shared with FastAPI ML service.  
Streamlit dashboard talks to the ML API. A daily ingestion worker pulls book mentions from four social platforms, scores sentiment, and writes :PlatformReception nodes into the same Neo4j.

— See *FedBookSem\_Architecture.drawio* for the full layered diagram.

## Layered Architecture — Five Tiers

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|                    |                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------|
| <b>Client</b>      | React FE (localhost:3000) · Streamlit dashboard (:8501) · remote ActivityPub servers               |
| <b>Application</b> | Express backend (:3001) · FastAPI ML service (:8000) · ActivityPub inbox / outbox / WebFinger      |
| <b>Ingestion</b>   | Reddit / YouTube / Bluesky / Mastodon collectors → entity resolver → sentiment scorer → aggregator |
| <b>Data</b>        | Neo4j 5.15 (bolt :7687) · Redis 7 (:6379) · PostgreSQL 16 (:5432) · cached model files on disk     |
| <b>External</b>    | Open Library · Hardcover GraphQL · HuggingFace Hub · Kaggle 7k · goodbooks-10k · remote AP peers   |

## Request Lifecycle — Login and Feed

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- Browser POSTs /api/auth/login → CRA dev proxy forwards to Express :3001.
- Express queries Neo4j MATCH (:Person {username}), bcrypt.compare, jwt.sign — returns JWT.
- Browser stores JWT in localStorage; axios interceptor attaches it as Bearer to every subsequent call.
- Browser GET /api/feed → Express verifies JWT → single Cypher: MATCH (me)-[:FOLLOWS]->(f)-[:AUTHORED]->(r)-[:REVIEWS]->(b).
- Response JSON hydrated on the client into a virtualised feed component.



## Federation Flow — Alice Follows @bob@remote.social

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- Alice clicks Follow → POST /api/social/follow with targetId.
- Backend WebFingers acct:bob@remote.social → discovers bob's actor URL.
- Fetches bob's actor JSON-LD, extracts publicKey and inbox URL.
- Constructs an ActivityPub Follow activity, signs it with alice's private key using HTTP Signatures (RSA-2048, SHA-256).
- POSTs the signed activity to bob's inbox; remote server verifies signature and MERGEs the follower.
- Remote server later sends an Accept activity back to alice's /inbox → verified → MERGE (alice)-[:FOLLOWS]->(bob).

## Recommender Request Flow — /recommend/similar

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- Client GET /recommend/similar?isbn=X&rerank;=ltr → FastAPI.
- FastAPI reads :Book(embedding) for X (or encodes free-text query with all-MiniLM-L6-v2).
- Neo4j native vector index returns top-50 candidates by cosine similarity.
- ReceptionScorer aggregates :PlatformReception nodes into per-platform sentiment.
- Optional cross-encoder (bge-reranker-base) or LightGBM LambdaRank fuses 6 features (sim, cf, reception, diversity, ce, freshness).
- Final ranked list returned with per-candidate breakdown — the same shape the dashboard renders.

## Ingestion Data Flow (privacy-preserving)

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- Scheduler triggers ingest\_daily.py (cron or manual).
- Query :Book WHERE description AND thumbnail exist LIMIT N (skips 6 Sri Lankan seed books — no English-language chatter).
- For each (book × collector): call platform API, extract mention text.
- Batch sentiment: twitter-roberta-base-sentiment-latest (batch\_size=16) → labels aggregated to (positive, neutral, negative).
- Persist only counts + external\_ids; raw mention text is discarded before writing.
- MERGE (b)-[:RECEPTION\_ON]->(r:PlatformReception) SET counts, ingested\_at, expires\_at (YouTube 30-day), demo=false.

## Neo4j Data Model — Nodes

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|                           |                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>:Person</b>            | id, username, displayName, bio, domain, publicKey, privateKey, passwordHash, avatarUrl, createdAt                   |
| <b>:Book</b>              | id, isbn, title, author, year, description, thumbnail, subjects, embedding (vector), goodbooksBookId, sourceCatalog |
| <b>:Review</b>            | id, content, rating, published, activityId, seedKey?                                                                |
| <b>:Annotation</b>        | id, motivation, bodyValue, exactText, prefix, suffix, startOffset, endOffset, bookSource, created, seedKey?         |
| <b>:PlatformReception</b> | platform, book_isbn, positive, neutral, negative, mentions, external_ids, ingested_at, expires_at, demo             |

## Neo4j Data Model — Relationships & Indexes

|                         |                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Social edges</b>     | <code>(:Person)-[:FOLLOWS]-&gt;(:Person) · (:Person)-[:LIKES]-&gt;(:Review)</code>                                                        |
| <b>Authoring</b>        | <code>(:Person)-[:AUTHORED]-&gt;(:Review) · (:Person)-[:ANNOTATED]-&gt;(:Annotation)</code>                                               |
| <b>Targets</b>          | <code>(:Review)-[:REVIEWS]-&gt;(:Book) · (:Annotation)-[:ON_SOURCE]-&gt;(:Book) · (:Book)-[:RECEPTION_ON]-&gt;(:PlatformReception)</code> |
| <b>Uniqueness</b>       | <code>CONSTRAINT :Person(username) UNIQUE · CONSTRAINT :Book(isbn) UNIQUE</code>                                                          |
| <b>Vector index</b>     | <code>bookEmbedding on :Book(embedding) — cosine, 384 dims (MiniLM-L6-v2)</code>                                                          |
| <b>Graph projection</b> | GDS projection over <code>:Person</code> → <code>:Person</code> for personalised PageRank                                                 |

# Deployment Topology

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|                      |                                                                             |
|----------------------|-----------------------------------------------------------------------------|
| <b>React FE</b>      | native · npm start · :3000 · CRA proxy → :3001                              |
| <b>Express BE</b>    | native · nodemon · :3001                                                    |
| <b>Neo4j 5.15</b>    | docker · :7474 (browser) + :7687 (bolt) · volumes: neo4j_data, neo4j_logs   |
| <b>Redis 7</b>       | docker · :6379                                                              |
| <b>PostgreSQL 16</b> | docker · :5432 · volume: postgres_data                                      |
| <b>ML service</b>    | docker (built) · :8000 · volumes: ml_hf_cache, ml_torch_cache, ./ml-service |
| <b>Dashboard</b>     | docker (built) · :8501 · same image as ml-service, streamlit entrypoint     |

## Recommender Strategies (all live)

|                                 |                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------|
| <b>Semantic</b>                 | MiniLM-L6-v2 embeddings + Neo4j native vector index (cosine k-NN)                  |
| <b>Cross-encoder rerank</b>     | BAAI/bge-reranker-base for query × candidate re-scoring                            |
| <b>Collaborative filtering</b>  | LightFM WARP trained on goodbooks-10k (~6M real ratings)                           |
| <b>Graph traversal</b>          | Neo4j GDS k-NN and personalised PageRank                                           |
| <b>Cross-platform reception</b> | Weighted sentiment across Reddit / YouTube / Bluesky / Mastodon                    |
| <b>Learned rank fusion</b>      | LightGBM LambdaRank over 6 features (sim, cf, reception, diversity, ce, freshness) |

# Cross-Platform Ingestion

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|          |                                                                                |
|----------|--------------------------------------------------------------------------------|
| Reddit   | PRAW OAuth · subreddits /r/books, /r/booksuggestions, /r/52book, /r/literature |
| YouTube  | Data API v3 · top-level comments on review videos · 30-day retention           |
| Bluesky  | AT Protocol public searchPosts · optional app-password auth                    |
| Mastodon | Public hashtag timelines · #bookstodon, #booksky, #booktok, #bookreview        |



## Federation — ActivityPub

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- Every user has an ActivityPub Person actor at `/users/:username`.
- WebFinger discovery at `/.well-known/webfinger` enables `acct:user@domain` lookup.
- HTTP Signatures (RSA-2048, node-forge) sign every outbound activity.
- Shared inbox at `/inbox` accepts Follow, Like, Announce, and Create activities.
- Reviews and annotations are exchanged as ActivityStreams objects with the W3C Anno JSON-LD context.

## Annotations — W3C Web Annotation

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- Full W3C Web Annotation Data Model with dual selectors — TextQuoteSelector + TextPositionSelector.
- Reader selects a passage; UI captures exact text, prefix/suffix, and character offsets.
- Annotation motivations supported: commenting, highlighting, tagging.
- Annotations propagate over ActivityPub as first-class Create activities.

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# Delivery

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## Timeline & Milestones

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- Slice A — federation core + annotations + React FE (done)
- Slice B — semantic recommender + LightFM CF + Neo4j vector index (done)
- Slice C — cross-platform ingestion + reception scoring + LTR fusion (done)
- Slice D — user study, evaluation, thesis writing (current)

## Reproducibility

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- One-command bring-up: `docker compose up -d` starts Neo4j, Postgres, Redis, ML service, dashboard.
- Idempotent seeders: `seed.js` (FE) and `seed_kaggle_books.py` (catalogue) both use MERGE, never wipe.
- Models are cached offline in the container (`HF_HUB_OFFLINE=1`) — no network needed after first build.
- All source, seeds, and configuration in one public git repository.

## Limitations & Future Work

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- LightFM is user-cold-start bound — needs onboarding taste-prompt to help new users.
- Cross-platform sentiment sample is small for less-discussed titles; needs longer accumulation window.
- Federation currently point-to-point; no relay support (BookWorm-style relay planned).
- Cross-encoder rerank is slow (~500 MB model); could be replaced with smaller distilled reranker.

# Thank You

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Questions?

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